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Information processing apparatus and control method thereof

Abstract

An information processing apparatus comprises: an arithmetic unit configured to perform arithmetic operation processing using a hierarchical network; a storage unit configured to store input data inputted to the arithmetic unit and output data outputted from the arithmetic unit; a transmission unit configured to transmit to the arithmetic unit the input data stored in the storage unit; a reception unit configured to receive and store in the storage unit the output data from the arithmetic unit; and a control unit configured to, in a case where the input data cannot be transmitted from the storage unit to the arithmetic unit, control supply of an operation clock to the transmission unit based on network information that indicates a structure of the hierarchical network.

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Background/Summary

BACKGROUND
Field of the Disclosure
(1) The present disclosure relates to power saving of an information processing apparatus.
Description of the Related Art
(2) Hierarchical arithmetic methods called CNNs (Convolutional Neural Networks) are attracting attention as methods that enable pattern recognition that is robust to variation in target objects. For example, Yann LeCun, Koray Kavukcuoglu and Clément Farabet, “Convolutional Networks and Applications in Vision”, Proc. International Symposium on Circuits and Systems (ISCAS'10), IEEE, 2010 discloses various examples of application and implementation. In order to reduce the power consumption of CNN processing

hardware, it is necessary to perform clock control in units of several small clock cycles.
(3) In an arithmetic operation apparatus, when stopping a clock supplied to a particular functional module that has an input or output data transfer interface, it is necessary to prevent the occurrence of inconsistencies with surrounding modules. In Japanese Patent Laid-Open No. 2003-58271, when a clock stop request signal is asserted, a function module completes the processing that is currently being performed. Japanese Patent Laid-Open No. 2003-58271 also discloses a method for reducing power in which a data transfer interface is transitioned to a state in which transfer cannot be performed, an acknowledge signal is returned, and then a clock that is being supplied is stopped. Furthermore, Japanese Patent No. 5751819 discloses a method of forcibly shutting off input and output in which the procedure of asserting a request signal and returning an acknowledge signal is omitted.
(4) However, in the method of Japanese Patent Laid-Open No. 2003-58271, there is a problem that it takes time for a procedure of asserting a request signal, returning an acknowledge signal, and then stopping a clock. Furthermore, in the method of Japanese Patent No. 5751819, since processing other than data transfer processing uniformly stops when a clock is stopped, processing that was being performed during a data transfer waiting time (such as next data transfer preparation) becomes impossible, and therefore, there is a possibility that the overall processing time will increase. Moreover, since a time lag occurs in the re-supply of a clock, there is a possibility that processing time may further increase.
(5) Accordingly, since the on/off of a clock is repeatedly performed even within CNN processing of one frame when it is desired to stop the clock with fine granularity, the power saving effect becomes low in the above-described method.
SUMMARY
(6) According to one aspect of the present disclosure, an information processing apparatus comprises: an arithmetic unit configured to perform arithmetic operation processing using a hierarchical network; a storage unit configured to store input data inputted to the arithmetic unit and output data outputted from the arithmetic unit; a transmission unit configured to transmit to the arithmetic unit the input data stored in the storage unit; a reception unit configured to receive and store in the storage unit the output data from the arithmetic unit; and a control unit configured to, in a case where the input data cannot be transmitted from the storage unit to the arithmetic unit, control supply of an operation clock to the transmission unit based on network information that indicates a structure of the hierarchical network.
(7) According to another aspect of the present disclosure, an information processing apparatus comprises: an arithmetic unit configured to perform arithmetic operation processing using a hierarchical network; a storage unit configured to store input data inputted to the arithmetic unit and output data outputted from the arithmetic unit; a transmission unit configured to transmit to the arithmetic unit the input data stored in the storage unit; a reception unit configured to receive and store in the storage unit the output data from the arithmetic unit; and a control unit configured to, in a case where it is not possible to transfer the output data, control supply of an operation clock to the reception unit based on network information that indicates a structure of the hierarchical network.
(8) The present disclosure efficiently reduces power consumption in an arithmetic operation apparatus.
(9) Further features of the present disclosure will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.
(2) FIG. **1** is a block diagram of a recognition processing unit according to one or more aspect of the present disclosure.
(3) FIG. **2** is a diagram illustrating a circuit configuration of a clock control unit.
(4) FIG. **3** is a diagram illustrating a timing chart of a circuit of the clock control unit.
(5) FIG. **4** is a diagram illustrating a two-wire handshake.
(6) FIG. **5** is a diagram illustrating a timing chart of the two-wire handshake.

(7) FIG. **6** is a diagram illustrating signals between a convolution operation unit **104** and a feature plane storage unit **102**.
(8) FIG. **7** is a flowchart for explaining overall operation of an image processing system.
(9) FIG. **8** is a flowchart for explaining detailed operation of the recognition processing unit.
(10) FIG. **9** is a flowchart of clock control.
(11) FIGS. **10**A and **10**B are diagrams illustrating a timing chart for explaining clock control at the time of feature plane generation.
(12) FIG. **11** is a diagram illustrating an overall configuration of the image processing system that uses the recognition processing unit.
(13) FIG. **12** is a diagram for explaining a network configuration of a CNN.
(14) FIG. **13** is a diagram for explaining a calculation operation of a feature plane **1205***a*.
(15) FIG. **14** is a diagram illustrating a time chart for explaining the calculation operation of the feature plane **1205***a*.

DESCRIPTION OF THE EMBODIMENTS

(16) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the disclosure. Multiple features are described in the embodiments, but limitation is not made to the disclosure that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

First Embodiment

(17) A description will be given using an image processing system as an example of a first embodiment of an information processing apparatus according to the disclosure. Hereinafter, operation for CNN processing, which serves as a basis, will be first described, and then an image processing system that has a recognition processing unit that performs such CNN processing will be described.
(18) <Convolutional Neural Network (CNN) Processing>
(19) FIG. **12** is a diagram for explaining an exemplary network configuration of a CNN. A CNN is constructed as a hierarchical network and is configured by an input layer **1201**, a first layer **1208**, a second layer **1209**, and a third layer **1210**. When performing CNN processing on image data, the input layer **1201** corresponds to image data of a predetermined size. Feature planes **1203***a* to **1203***d* are feature planes of the first layer **1208**, feature planes **1205***a* to **1205***d* are feature planes of the second layer **1209**, and a feature plane **1207** is a feature plane of the third layer **1210**. A feature plane is a data plane corresponding to a result of processing of a predetermined feature extraction operation (convolution operation and non-linear processing). The feature plane corresponds to a result of feature extraction for recognizing a predetermined object in an upper layer, and since it is a result of processing on image data, the result of the processing is also represented by a plane.
(20) The feature planes **1203***a* to **1203***d* are generated by a convolution operation and non-linear processing corresponding to the input layer **1201**. For example, the feature plane **1203***a* is calculated by a two-dimensional convolution operation schematically illustrated in a convolution kernel **12021***a* and a non-linear transformation of the arithmetic operation result. For example, the convolution operation of a kernel (coefficient matrix) size of columnSize×rowSize is a product-sum operation as indicated below.
(21)

$$\text{output}(x,y)=\sum_{l=1}^{L}\ \sum_{\text{row}=-\text{rowSize}/2}^{\text{rowSize}/2}\ \sum_{\text{column}=-\text{columnSize}/2}^{\text{columnSize}/2}\text{input}(x+\text{column},y+\text{row})\times\text{weight}(\text{column},\text{row})\qquad(1)$$

(22) Here, input(x, y): reference pixel value at coordinates (x, y) output(x, y): arithmetic operation result at coordinates (x, y) weight(column, row): weighting coefficient used for calculating output(x, y) columnSize, rowSize: convolution kernel size L: number of feature maps in the previous layer.
(23) In CNN processing, a feature plane is calculated by repeating a product-sum operation while scanning a plurality of convolution kernels in pixel units and performing non-linear transformation on the final product-sum result. In the case of calculating the feature plane **1203***a*, since the number of connections with the previous layer is "1", there is one convolution kernel. Here, the convolution kernels **12021***b*, **12021***c*, and **12021***d* are convolution kernels used when calculating the feature planes **1203***b*, **1203***c*, and **1203***d*, respectively.
(24) FIG. **13** is a diagram for explaining the operation for calculating the feature plane **1205***a*. As illustrated

in FIG. **12**, the feature plane **1205***a* is connected to the four feature planes **1203***a* to *d* of the first layer **1208**, which is the previous layer. In the case of calculating the data of the feature plane **1205***a*, first, a filter operation unit **1301** performs a filter operation on the feature plane **1203***a* using a kernel **12041***a* and holds the result in a cumulative adder **1302**. Similarly, the feature planes **1203***b*, **1203***c*, and **1203***d* are subjected to a convolution operation using kernels **12042***a*, **12043***a*, and **12044***a*, respectively, and the results are accumulated in the cumulative adder **1302**. A non-linear transformation processing unit **1303** performs non-linear transformation processing using a logistic function or a hyperbolic tangent function (tanh function) after the completion of four types of convolution operation. The feature plane **1205***a* is calculated by performing the above processing while scanning the entire image pixel by pixel. Similarly, the feature plane **1205***b* is calculated using four convolution operations using the kernels **12041***b*, **12042***b*, **12043***b*, and **12044***b* with respect to the four feature planes **1203***a* to *d* of the first layer **1208** which is the previous layer.

(25) Furthermore, the feature plane **1207** is calculated using four convolution operations using the kernels **1261**, **1262**, **1263**, and **1264** with respect to the four feature planes **1205***a* to *d* of the second layer **1209** which is the previous layer. It is assumed that each kernel coefficient is determined in advance by learning using general methods such as perceptron learning and back propagation learning.

(26) When the CNN processing hardware for performing the CNN operation described above is implemented in an embedded system to perform network processing, the CNN processing hardware performs an arithmetic operation of input data and weighting coefficients for each layer. Then, an arithmetic operation result is made to be input data of the next layer, and an arithmetic operation with weighting coefficients of the next layer is repeated to obtain the final pattern recognition result.

(27) The CNN processing needs an enormous number of product-sum operations to repeat a large number of convolution operations, and it is necessary for the CNN processing hardware to perform processing at high speed. In addition, a large-scale internal memory is also necessary to process a large-scale network, and power consumption becomes a problem.

(28) As a method of achieving power saving without lowering processing speed, it is conceived that an arithmetic unit is always operated, and regarding the arithmetic unit and a data transfer control unit for performing input/output data transfer, a clock is controlled to turn on/off in accordance with data transmission and reception operation of the arithmetic unit.

(29) FIG. **14** is a diagram illustrating a time chart for explaining the operation for calculating the feature plane **1205***a*. In the drawing, the processing being performed by each of the filter operation unit **1301**, the cumulative adder **1302**, and the non-linear transformation processing unit **1303** is illustrated. Here, it is assumed that there is one filter operation unit **1301** in the CNN processing hardware and it performs filter operations on feature planes in a time division manner. Furthermore, it is assumed that the filter operation unit **1301** takes one cycle for a product-sum operation of one pixel.

(30) The filter operation unit **1301** performs a filter operation on the feature plane **1203***a* using the kernel **12041***a* (processing **1401**). When the kernel size is columnSize×rowSize, the product-sum operation is performed columnSize×rowSize times. Therefore, columnSize×rowSize cycles are necessary for processing. The cumulative adder **1302** sequentially performs cumulative addition of a result of calculation by the filter operation unit **1301** (processing **1402**).

(31) The filter operation unit **1301** continues to sequentially perform filter operations using the kernels **12042***a* to **12044***a* for the feature planes **1203***b* to *d*, and the cumulative adder **1302** sequentially performs cumulative addition. When the result of the filter operation on the last feature plane of the previous layer is added, the non-linear transformation processing unit **1303** performs processing (processing **1403**) to obtain the feature plane **1205***a*.

(32) As described above, the feature plane of one pixel is obtained once in columnSize×rowSize×L cycles. Therefore, a clock need only be supplied to the arithmetic unit and the data transfer control unit for transferring input/output data only during a period in which it is necessary. Therefore, it is possible to expect efficient power saving by stopping and supplying a clock in cycle units of fine granularity.

(33) <System Configuration>

(34) FIG. **11** is a diagram illustrating an overall configuration of the image processing system that uses the recognition processing unit. The system has a function of detecting a region of a specific object from image data by performing pattern recognition.

(35) An image input unit **1100** is a functional unit that inputs image data to be processed. The image input

unit **1100** is, for example, an image capturing device and is configured by an optical system, a driver circuit for controlling a photoelectric conversion device and a sensor, an AD converter, a signal processing circuit for controlling various kinds of image correction, a frame buffer, and the like. The photoelectric conversion device and the sensor include CCDs (Charge-Coupled Devices), CMOSs (Complementary Metal Oxide Semiconductors), and the like.

(36) A recognition processing unit **1101** is a functional unit that performs pattern recognition. Details of the recognition processing unit **1101** will be described later with reference to FIG. **1**. A DMAC (Direct Memory Access Controller) **1105** is responsible for transferring data between each functional unit connected to an image bus **1102** and a CPU bus **1109**. A bridge **1103** provides a bridge function for the image bus **1102** and the CPU bus **1109**. A preprocessing unit **1104** performs various kinds of preprocessing for effectively performing pattern recognition processing. Specifically, image data conversion processing such as color conversion processing and contrast correction processing is processed by hardware.

(37) A CPU **1106** controls the operation of the entire image processing system. A ROM (Read Only Memory) **1107** stores instructions and operation parameter data that define the operation of the CPU **1106** and a data set for operating the recognition processing unit **1101**. Here, the data set includes operation parameters and weight coefficients that accord with the neural network. The data set is inputted to the recognition processing unit **1101** via the DMAC **1105**. A RAM **1108** is a working memory that is necessary for the operation of the CPU **1106**, for example.

(38) Image data inputted from the image input unit **1100** is processed by the preprocessing unit **1104** and temporarily stored in the RAM **1108**. Thereafter, the image data is inputted from the RAM **1108** to the recognition processing unit **1101** via the DMAC **1105**. When a data set is received, the recognition processing unit **1101** performs predetermined determination processing on the inputted preprocessed image data in pixel units to determine a region of a predetermined object in the input image. The result of the determination is stored in the RAM **1108** via the DMAC **1105**.

(39) <Configuration of Recognition Processing Unit>

(40) FIG. **1** is a block diagram of the recognition processing unit **1101** according to the first embodiment. A control unit **101** controls the entire recognition processing unit **1101**. Furthermore, the control unit **101** includes a network information holding unit **1011** for holding network information. As described above, the network information is set by the DMAC **1105** transferring the operation parameters placed in advance in the ROM **1107**. The network information includes, for example, the number of layers to be processed, a layer number, information on feature planes for each layer (the width and height of a feature plane, the number of feature planes, and the like), and information on connections between layers (the horizontal size and the vertical size of a convolution kernel and the like).

(41) The control unit **101** also includes an enable register as a register for operation control, and the CPU **1106** instructs a start of processing. The CPU **1106** instructs the control unit **101** to start processing of a plurality of layers, and the control unit **101** instructs the feature plane storage unit **102** and the convolution operation unit **104** to be described later to start processing in layer units a plurality of times. The control unit **101** transmits, as a control parameter, data that has been set in the network information holding unit **1011** together with a control signal that indicates a processing start instruction. When performing post-processing on the generated feature planes of a layer, the control unit **101** transmits, as a control parameter, the data that has been set in the network information holding unit **1011** to the CPU **105** after the processing in layer units has been completed in the feature plane storage unit **102** and the convolution operation unit **104**. Then, an instruction to start post-processing is issued.

(42) The feature plane storage unit **102** has a feature plane holding unit **1021** for storing feature planes. Furthermore, the feature plane storage unit **102** includes a data transmission unit **1022** that reads a feature plane from the feature plane holding unit **1021** and transmits it to the convolution operation unit **104** and a data reception unit **1023** that receives an arithmetic operation result of the convolution operation unit **104** and stores it in the feature plane holding unit **1021**.

(43) The feature plane holding unit **1021** is configured by, for example, a dual port SRAM, so that reading by the data transmission unit **1022** and storage by the data reception unit **1023** can be performed simultaneously.

(44) The data transmission unit **1022** reads and transmits a feature plane referenced by the convolution operation unit **104** from the feature plane holding unit **1021** based on a control parameter (that is, hierarchical network information) received from the control unit **101**. A clock is supplied by a clock control

unit **103** to be described later. The data transmission unit **1022** instructs the clock control unit **103** to stop the clock and gives it a count value for controlling an interval from the stop to re-supply of the clock.
(45) The data reception unit **1023** receives as a feature plane an arithmetic operation result outputted by the convolution operation unit **104** based on a control parameter (that is, hierarchical network information) received from the control unit **101** and then stores it in the feature plane holding unit **1021**. When all the feature planes are received, the data reception unit **1023** notifies the control unit **101** of the completion. A clock is supplied by the clock control unit **103** to be described later. The data reception unit **1023** instructs the clock control unit **103** to stop the clock and gives it a count value for controlling an interval from the stop to re-supply of the clock (that is, a period during which the supply is stopped).
(46) The clock control unit **103** supplies operation clocks to the data transmission unit **1022** and the data reception unit **1023**. The clock control unit **103** includes a reception timer **1032** used for clock control of the data reception unit **1023** and a transmission timer **1031** used for clock control of the data transmission unit **1022**.
(47) The clock control unit **103** stops supplying a clock when a stop instruction and a count value are received from the data transmission unit **1022** and starts counting with the transmission timer **1031**. The clock control unit **103** re-supplies the clock once it has counted to the count value received from the data transmission unit **1022** and sets the transmission timer **1031** to an initial state.
(48) Similarly, the clock control unit **103** stops supplying a clock when a stop instruction and a count value are received from the data reception unit **1023** and starts counting with the reception timer **1032**. The clock control unit **103** re-supplies the clock once it has counted to the count value received from the data reception unit **1023** (the count value received from the data reception unit **1023** has expired) and sets the reception timer **1032** to an initial state.
(49) FIG. **2** is a diagram illustrating a circuit configuration of the clock control unit **103**. A circuit **201***a* is a circuit for controlling the clock of the data transmission unit **1022**. A circuit **201***b* is a circuit for controlling the clock of the data reception unit **1023**. Since the configuration and operation of the circuit **201***a* and the circuit **201***b* are substantially the same, in the following, a description will be made only on the circuit **201***a*.
(50) The transmission timer **1031** starts counting when a stop instruction pulse signal (stop) is received from the data transmission unit **1022**. When the count reaches the count value (val) specified by the data transmission unit **1022**, a count completion pulse signal (cnt_done) is asserted, the counting is stopped, and the count is returned to an initial value.
(51) A clock enable generation circuit **202** disables a clock enable signal (clk_en) when a stop is received from the data transmission unit **1022**. Furthermore, the clock enable generation circuit **202** enables a clock enable signal (clk_en) when cnt_done is received from the transmission timer **1031**. A circuit **203** is a clock gating circuit of a general configuration and turns on the clock when clk_en is 1 and turns off the clock when clk_en is 0.
(52) FIG. **3** is a diagram illustrating a timing chart of the circuit **201***a* of the clock control unit **103**. The transmission timer **1031** sets the stop instruction signal stop to “1” at t**0** and the count value val is set to, for example, “5”. The transmission timer **1031** increments an internal counter value counter every cycle starting from t**1** and the internal counter value counter matches val “5” at t**4**. Therefore, the transmission timer **1031** asserts ent_done at t**4**. Then, the transmission timer **1031** sets a counter value counter to an initial state at t**5** and deasserts cnt_done.
(53) The clock enable generation circuit **202** sets clk_en to “0” at t**1** when stop is asserted. When clk_en is set to “0”, a gated clock gclk stops the clock in the next cycle (that is, t**2**). Moreover, when cnt_done is asserted at t**4**, clk_en is set to “1” at t**5**, and gclk re-supplies the clock at the next cycle (that is, t**6**). A clock stop cycle has five cycles (from t**2** to t**5**) in accordance with the count value val. That is, the clock stop cycle can be controlled by the count value val.
(54) The convolution operation unit **104** is a functional unit for performing a convolution operation. Weighting coefficients that accord with a kernel size and a reference feature plane are taken as input, and a feature plane is outputted as an arithmetic operation result. Here, it is assumed that the DMAC **1105** transfers to the convolution operation unit **104** the weighting coefficients placed in advance in the ROM **1107**. The convolution operation unit **104** receives a kernel size as a control parameter from the control unit **101** and, when instructed to start processing, performs convolution operation processing, cumulative addition processing, and non-linear transformation processing. When the arithmetic operation processing of

all the feature planes of a layer to be generated is completed, the completion is notified to the control unit **101**. As described above, in the present embodiment, it is assumed that it takes one cycle for a convolution operation of one pixel. Therefore, it takes columnSize×rowSize cycles to process a feature plane with a kernel size columnSize×rowSize.
(55) A two-wire handshake can be utilized for data transfer between the convolution operation unit **104** and the feature plane storage unit **102**. FIG. **4** is a diagram illustrating data transmission by a two-wire handshake.
(56) A data transmitting side **401** sets data to be transmitted to a data line Data when it is ready for data transmission and sets a Valid signal that indicates that data transmission is in progress to "1". A data receiving side **402** sets a Ready signal that indicates that reception is possible to "1" when it is ready to receive data. Data is transferred when both the Valid signal and the Ready signal are "1".
(57) FIG. **5** is a diagram illustrating a timing chart of the two-wire handshake. The data transmitting side **401** starts transmitting data DO at t**0**. Ready of the data receiving side **402** is "1", and the transfer of the data DO is completed at t**1**. The data transmitting side **401** further attempts to transmit the next data D**1**, but since Ready of the data receiving side **402** is "0", the transfer is unsuccessful at t**2**, and the data transmitting side **401** further continues data transmission. The data receiving side **402** sets Ready to "1" with the clock at t**2**, and the transfer of the data D**1** is completed at **3**.
(58) FIG. **6** is a diagram illustrating signals between the convolution operation unit **104** and the feature plane storage unit **102**. Here, each of the data transmission unit **1022** and the data reception unit **1023** in the feature plane storage unit **102** performs data transmission by a two-wire handshake with the convolution operation unit **104**.
(59) That is, in the data transfer of a reference feature plane from the data transmission unit **1022** to the convolution operation unit **104**, the data transmission unit **1022** controls a Valid signal (ref_Valid) and a Data signal (ref_Data). Furthermore, the convolution operation unit **104** controls a Ready signal (ref_Ready). The data width of ref_Data is a data width for reference pixels (columnSize×rowSize pixels) that are necessary for a convolution operation of the kernel size columnSize×rowSize to be capable of being transferred in one cycle.
(60) Furthermore, in the data transfer of an arithmetic operation result from the convolution operation unit **104** to the data reception unit **1023**, the convolution operation unit **104** controls a Valid signal (calc_Valid) and a Data signal (calc_Data). Moreover, the data reception unit **1023** controls a Ready signal (calc_Ready). The data width of calc_Data is a data width for one pixel of a feature plane to be capable of being transferred in one cycle.
(61) A CPU **105** performs the post-processing of a feature plane. The CPU **105** is a bus master of the image bus **1102**, and in a memory space, a memory of the feature plane holding unit **1021** is address-mapped via a memory control unit **106** to be described later. The CPU **105** acquires data when a processing start instruction is received from the control unit **101** and performs processing.
(62) The memory control unit **106** controls reading and writing of the memory of the feature plane holding unit **1021**. The memory control unit **106** is a bus slave of the image bus **1102**, receives a request from the bus master, and reads and writes the feature plane holding unit **1021**.
(63) <Apparatus Operation>
(64) FIG. **7** is a flowchart for explaining the overall operation of the image processing system. The following operation is performed, for example, by the CPU **1106** executing a control program.
(65) In step S**701**, the CPU **1106** starts processing of the system. In step S**702**, the CPU **1106** acquires input data via the image input unit **1100**. In step S**703**, the CPU **1106** performs preprocessing on the acquired data via the preprocessing unit **1104** and stores the result in the RAM **1108**.
(66) In step S**704**, the CPU **1106** sets a start position address of a data set of a layer to be processed placed in the ROM **1107** to the DMAC **1105** and activates the DMAC **1105**. One or more layers may be processed. In step S**705**, the CPU **1106** sets a start position address of preprocessed data in the RAM **1108** to the DMAC **1105** activates the DMAC **1105**.
(67) In step S**706**, the CPU **1106** activates the recognition processing unit **1101** and performs recognition processing. When the processing of the recognition processing unit **1101** is completed, the processing result (a detection result or feature data of a middle layer) is stored in the RAM **1108**.
(68) In step S**707**, the CPU **1106** determines whether or not processing of all the layers is completed. When the processing of all layers is completed (Yes), the processing is terminated, and when the processing is not

completed (No), the processing returns to step S**704**. For example, in a small network for embedded devices, it is possible to process all the layers at once, and therefore, it may be Yes in step S**707** of the first loop. Meanwhile, in the processing of a large neural network, all the layers cannot be processed at once, and therefore, the network is processed in a time division manner. In such a case, it becomes No in step S**707** of the first loop, the processing returns to step S**704** again, and the processing of the remaining layers is performed. The processing in step S**705** in the second and subsequent loops sets the processing result of the recognition processing unit **1101** stored in the RAM **1108** as the processing target.
(69) FIG. **8** is a flowchart illustrating detailed operation (step S**706**) of the recognition processing unit **1101**. The following operation of each unit in the recognition processing unit **1101** is controlled by, for example, the control unit **101**. At the time of completion of step S**705**, information on the layer that is a processing target of the recognition processing unit **1101** is set in the network information holding unit **1011**. In step S**801**, the recognition processing unit **1101** starts processing.
(70) In step S**802**, the control unit **101** transmits the information held in the network information holding unit **1011** to the feature plane storage unit **102** and the convolution operation unit **104** and issues a processing start instruction. Here, it is assumed that the information held in the network information holding unit **1011** is transmitted as a control parameter in units of layers.
(71) In step S**803**, the feature plane storage unit **102** determines whether or not it is the processing of an input layer. If it is the processing of an input layer (Yes), the processing proceeds to step S**804**, and if it is not the processing of an input layer (No), the processing proceeds to step S**805**. In step S**804**, the data transmission unit **1022** outputs image data that is input data as a reference feature plane. In step S**805**, the data transmission unit **1022** reads a feature plane of the previous layer from the feature plane holding unit **1021** and outputs it as a reference feature plane.
(72) In step S**806**, the convolution operation unit **104** performs a convolution operation based on the reference feature plane and weight coefficients and transmits an arithmetic operation result to the feature plane storage unit **102**. In step S**807**, the data reception unit **1023** of the feature plane storage unit **102** stores the arithmetic operation result in the feature plane holding unit **1021**.
(73) In step S**808**, the feature plane storage unit **102** determines whether or not generation of all the feature planes is completed. When the generation of all the feature planes is completed (Yes), the feature plane storage unit **102** and the convolution operation unit **104** issues a completion notification to the control unit **101** and proceeds to step S**809**. If not completed (No), the processing returns to step S**803** and the next feature plane is generated.
(74) In step S**809**, when the completion notification is received, the control unit **101** determines whether or not the processing of the final layer is completed. If the processing of the final layer is completed (Yes), the processing proceeds to step S**810**. If the processing of the final layer is not completed (No), the control unit **101** instructs the processing of the next layer and returns to step S**802**.
(75) In step S**810**, the control unit **101** notifies the network information such as a final layer number to the CPU **105**. Based on the network information, the CPU **105** reads the feature plane data of the final layer and specifies position coordinates.
(76) FIG. **9** is a flowchart of clock control. In the following, the operation in data transmission from the data transmission unit **1022** to the convolution operation unit **104** and the operation in data transmission from the convolution operation unit **104** to the data reception unit **1023** will be described with reference to FIG. **9**.
(77) First, the clock control of the data transmission unit **1022** in a step (step S**805**) in which the data transmission unit **1022** reads a feature plane of the previous layer from the feature plane holding unit **1021** and outputs it as a reference feature plane will be described. As described above, step S**805** is the processing of outputting a reference feature plane of the previous layer in the processing of generating one feature plane. When there are a plurality of feature planes of the previous layer, a plurality of planes are repeatedly outputted. Data transfer is performed using the ref_Valid, ref_Data, and ref_Ready signals illustrated in FIG. **6**.
(78) In step S**901**, the data transmission unit **1022** starts processing. In step S**902**, the data transmission unit **1022** asserts ref_Valid and starts data transfer of the reference feature plane. In step S**903**, the data transmission unit **1022** determines whether the transfer is successful. If ref_Ready is “1”, it is determined that the transfer is successful, and the processing proceeds to step S**904**. Meanwhile, if it is “0”, it is determined that the transfer is not successful, and the processing proceeds to step S**905**. In step S**904**, the

data transmission unit **1022** determines whether the transfer of the final data is completed. If the transfer of the final data is completed (Yes), the processing is terminated. If the transfer of the final data is not completed (No), the processing returns to step S**902** and data transfer is continued.
(79) In step S**905**, the data transmission unit **1022** determines whether to implement clock control based on a clock control implementation criterion to be described later. If it is determined to not carry out implementation (No), the processing returns to step S**902** and data transfer is continued. If it is determined to carry out implementation (Yes), the processing proceeds to step S**906**.
(80) In step S**906**, the data transmission unit **1022** sets the transfer signal ref_Valid to "0", and transition is made to a state in which no further transfer is performed. In step S**907**, the data transmission unit **1022** sets a count value to be transferred to the clock control unit **103** and issues a clock stop instruction. In step S**908**, the data transmission unit **1022** waits for the re-supply of the clock.
(81) In step S**909**, when the clock control unit **103** receives the clock stop instruction, the transmission timer **1031** starts counting. Furthermore, the clock enable generation circuit **202** disables the clock enable signal and stops supplying the clock.
(82) In step S**910**, the transmission timer **1031** continues to count until a count value specified by the data transmission unit **1022** is reached. In step S**911**, the transmission timer **1031** stops counting when the count value is reached and asserts the count completion pulse signal (cnt_done). Furthermore, the clock enable generation circuit **202** enables the clock enable signal and re-supplies the clock.
(83) In step S**912**, after the re-supply of the clock, the data transmission unit **1022** starts data transfer and waits for the transfer to be successful. When the transfer is successful, the processing proceeds to step S**904**, and it is determined whether or not the transfer of the final data is completed.
(84) As the clock control implementation criterion used in the determination in step S**905** described above, it is possible to use a comparison between the number of time lag cycles and the number of processing cycles. Here, the number of time lag cycles is the sum of the number of cycles from when it is determined that the transfer has failed in step S**903** until the clock supply is stopped and the number of cycles from when the count value is reached in step S**910** until the clock is re-supplied and the transfer is started. Furthermore, the number of processing cycles is the number of cycles it takes for the convolution operation unit **104** to process one reference feature plane and then request for the next reference feature plane. When the number of processing cycles exceeds the number of time lag cycles T, it is assumed that the clock is controllable without processing time overhead and "clock control is implemented".
(85) For example, assuming that the convolution kernel size is columnSize×rowSize, when the number of processing cycles is columnSize×rowSize>T, clock control is implemented. The number of time lag cycles T may include a cycle of processing that has been performed during the data transfer waiting time such as processing for the next data transfer preparation that is performed after the data transfer is successful. In order to reduce the processing time overhead and achieve better power saving effect, columnSize×rowSize−T should be set as the count value in step S**907**.
(86) Next, the clock control of the data reception unit **1023** in the steps in which the convolution operation unit **104** transmits the arithmetic operation result to the data reception unit **1023** (step S**806**) and the data reception unit **1023** stores the arithmetic operation result in the feature plane holding unit **1021** (step S**807**) will be described. As described above, step S**806** is the processing of outputting a cumulative addition result of all the feature planes of the previous layer in the processing of generating one feature plane. Data transfer is performed using the calc_Valid, calc_Data, and calc_Ready signals illustrated in FIG. **6**.
(87) In step S**901**, the data reception unit **1023** starts processing. In step S**902**, the data reception unit **1023** asserts calc_Ready and waits for data transfer of the arithmetic operation result. In step S**903**, the data reception unit **1023** determines whether the transfer is successful. If calc_Valid is "1", it is determined that the transfer is successful, and the processing proceeds to step S**904**. Meanwhile, if it is "0", it is determined that the transfer is not successful, and the processing proceeds to step S**905**. In step S**904**, the data reception unit **1023** determines whether the transfer of the final data is completed. If the transfer of the final data is completed (Yes), the processing is terminated. If the transfer of the final data is not completed (No), the processing returns to step S**902** and data transfer is continued.
(88) In step S**905**, the data reception unit **1023** determines whether to implement clock control based on a clock control implementation criterion to be described later. If it is determined to not carry out implementation (No), the processing returns to step S**902** and data transfer is continued. If it is determined to carry out implementation (Yes), the processing proceeds to step S**906**.

(89) In step S**906**, the data reception unit **1023** sets the transfer signal calc_Ready to "0", and transition is made to a state in which no further transfer is performed. In step S**907**, the data reception unit **1023** sets a count value to be transferred to the clock control unit **103** and issues a clock stop instruction. In step S**908**, the data reception unit **1023** waits for the re-supply of the clock.
(90) In step S**909**, when the clock control unit **103** receives the clock stop instruction, the reception timer **1032** starts counting. Furthermore, the clock enable generation circuit **202** disables the clock enable signal and stops supplying the clock.
(91) In step S**910**, the reception timer **1032** continues to count until a count value specified by the data reception unit **1023** is reached. In step S**911**, the reception timer **1032** stops counting when the count value is reached and asserts the count completion pulse signal (cnt_done). Furthermore, the clock enable generation circuit **202** enables the clock enable signal and re-supplies the clock.
(92) In step S**912**, after the re-supply of the clock, the data reception unit **1023** starts data transfer and waits for the transfer to be successful. When the transfer is successful, the processing proceeds to step S**904**, and it is determined whether or not the transfer of the final data is completed.
(93) As the clock control implementation criterion used in the determination in step S**905** described above, it is possible to use a comparison between the number of time lag cycles and the number of processing cycles. Here, the number of time lag cycles is the sum of the number of cycles from when it is determined that the transfer has failed in step S**903** until the clock supply is stopped and the number of cycles from when the count value is reached in step S**910** until the clock is re-supplied and the transfer is started. Furthermore, the number of processing cycles is the number of cycles it takes for the convolution operation unit **104** to generate one feature plane until generation of the next feature plane. When the number of processing cycles exceeds the number of time lag cycles T, it is assumed that the clock is controllable without processing time overhead and "clock control is implemented".
(94) For example, if the convolution kernel size is columnSize×rowSize and the number of feature planes of the previous layer is L, the clock control is implemented when the number of processing cycles is L×columnSize×rowSize>T. The number of time lag cycles T may include a cycle of processing that has been performed during the data transfer waiting time such as processing for the next data transfer preparation that is performed after the data transfer is successful. In order to reduce the processing time overhead and achieve better power saving effect, L×columnSize×rowSize−T should be set as the count value in step S**907**.
(95) FIGS. **10**A and **10**B are diagrams illustrating a timing chart for explaining clock control at the time of feature plane generation. More specifically, it is a timing chart illustrating the operation of clock control of the data transmission unit **1022** and the data reception unit **1023** for when the feature plane **1205***a* of FIG. **13** is generated in FIG. **1**. The convolution kernel size of the feature plane **1205***a* is 3×3. The feature planes **1203***a*, **1203***b*, **1203***c*, and **1203***d* are sequentially inputted and processed to generate the feature plane **1205***a*.
(96) In the present embodiment, the number of cycles from when it is determined that the transfer has failed in step S**903** until the clock supply is stopped is "2" cycles and the number of cycles from when the count value is reached in step S**910** until the clock is re-supplied and the transfer is started is "2" cycles. That is, the number of time lag cycles is "4" cycles.
(97) At t**0**, when the feature plane storage unit **102** receives a processing start instruction (step S**802**), the data transmission unit **1022** starts data transfer. At t**0**, the data transmission unit **1022** asserts ref_Valid (step S**902**) and the convolution operation unit **104** asserts ref_Ready, and thereby, the transfer of the feature plane **1203***a* is successful.
(98) At t**1**, the convolution operation unit **104** starts processing using the transferred data. When the convolution operation unit **104** negates ref_Ready, transfer fails (step S**903**), and clock control implementation is determined (step S**905**). Here, the number of processing cycles is 3×3=9 cycles and the number of time lag cycles is 4 cycles, and therefore, it is determined to "implement clock control".
(99) At t**2**, the data transmission unit **1022** negates ref_Valid (step S**906**), asserts the clock stop instruction stop, and issues a clock stop instruction (step S**907**). At this time, the data transmission unit **1022** provides 3×3-4=5 as the counter value to the transmission timer **1031**.
(100) At t**3**, clk_en=0 due to the clock stop instruction, and the clock of the data transmission unit **1022** is stopped. The transmission timer **1031** starts counting, reaches the count value at t**4**, and asserts the count completion pulse signal cnt_done.

(101) At t**5**, clk_en=1 due to the count completion pulse signal, and the clock of the data transmission unit **1022** is re-supplied (step S**911**). When the clock is re-supplied, at t**6**, ref_Valid is asserted and the transfer of the feature plane **1203***b* is started (step S**912**).
(102) Similarly to the transfer of the feature plane **1203***b* described above, the feature planes **1203***c* and **1203***d* are also transferred while the clock is turned on and off.
(103) Furthermore, at t**0**, when the feature plane storage unit **102** receives a processing start instruction (step S**802**), the data reception unit **1023** also starts data reception. At t**0**, the data reception unit **1023** asserts calc_Ready (step S**902**), and, since there is yet to be an arithmetic operation result at the convolution operation unit **104**, the data reception unit **1023** deasserts calc_Valid. The transfer fails (step S**903**), and implementation of clock control is determined (step S**905**). Here, the number of processing cycles is 4×3 3=36 cycles and the number of time lag cycles is 4 cycles, and therefore, it is determined to “implement clock control”.
(104) At t**1**, the data reception unit **1023** negates calc_Ready (step S**906**), asserts the clock stop instruction stop, and issues a clock stop instruction (step S**907**). At this time, the data reception unit **1023** provides 4×3×3−4=32 as the counter value to the reception timer **1032**.
(105) At t**2**, clk_en=0 due to the clock stop instruction, and the clock of the data reception unit **1023** is stopped. The reception timer **1032** starts counting, reaches the count value at t**7**, and asserts the count completion pulse signal cnt_done.
(106) At t**8**, clk_en=1 due to the count completion pulse signal, and the clock of the data reception unit **1023** is re-supplied (step S**911**). When the clock is re-supplied, at t**9**, calc_Ready is asserted and the reception of the feature plane **1205***a* is started (step S**912**).
(107) At t**10**, the convolution operation unit **104** asserts calc_Valid and outputs data, and the data transfer of the feature plane **1205***a* is completed.
(108) According to the first embodiment as described above, clock control is performed in the data transfer from the data transmission unit **1022** to the convolution operation unit **104**, and in the data transfer from the convolution operation unit **104** to the data reception unit **1023**. More specifically, when the transfer fails, an interface is transitioned to a state in which transfer is not performed and a clock stop instruction is issued to the clock control unit **103**. By this, it becomes unnecessary to perform a time-consuming procedure, and it becomes possible to obtain a better power saving effect in the data transmission unit **1022** and the data reception unit **1023**.
(109) Furthermore, when a clock stop instruction is issued to the clock control unit **103**, by providing a count value that accords with the network information of the convolution operation unit **104**, the clock control unit **103** can re-supply the clock at a suitable timing. That is, by re-supplying the clock based on a provided count value and resuming the data transfer processing, the clock control unit **103** can reduce processing time overhead.
(110) In the present embodiment, a description that a power saving effect is obtained with respect to data transfer has been given; however, the effect of reducing the power consumption of a memory is also obtained. That is, it also becomes possible to cope with the problem that, in a large-scale network, memory capacity increases and the power consumption by a memory increases. That is, by connecting the clock of the data transmission unit **1022** to a memory for reading transmission data or connecting the clock of the data reception unit **1023** to a memory for storing received data, a better power saving effect is obtained.
(111) The unit to which the clock is connected is not limited to a memory. That is, the clock of the data transmission unit **1022** may be connected to another unit involved in data transmission and the clock of the data reception unit **1023** may be connected to another unit involved in data reception.
(112) <Variation>
(113) Although in the above-described embodiment the description has been given on the case of using a two-wire handshake for data transfer, another data transfer may be utilized. For example, the present disclosure can be applied to a protocol that responds to a transfer request.
(114) Furthermore, in the above embodiment, the description has been given on the case of the CNN operation processing as the feature extraction processing; however, another arithmetic operation processing may be utilized. For example, the present disclosure can be applied to other various hierarchical processing such as perceptrons.
Other Embodiments
(115) Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus

that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

(116) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(117) This application claims the benefit of Japanese Patent Application No. 2021-113135, filed Jul. 7, 2021 which is hereby incorporated by reference herein in its entirety.

Claims

1. An information processing apparatus comprising: one or more memories storing instructions; and one or more processors executing the instructions to cause the information processing apparatus to function as: an arithmetic unit that performs arithmetic operation processing using a hierarchical network; a storage unit that stores input data inputted to the arithmetic unit and output data outputted from the arithmetic unit; a transmission unit that transmits to the arithmetic unit the input data stored in the storage unit; a reception unit that receives and stores in the storage unit the output data from the arithmetic unit; and a control unit that controls, in a case where the input data cannot be transmitted from the storage unit to the arithmetic unit, supply of an operation clock to the transmission unit based on network information that indicates a structure of the hierarchical network, wherein the hierarchical network is a CNN (Convolutional Neural Network), wherein the arithmetic operation processing includes convolution operation processing on the input data, the network information includes at least one of a size of a convolution kernel that is used in the convolution operation processing on the input data and the number of feature planes, wherein the control unit includes: a determination unit that determines, in a case where it is not possible to transfer the input data, whether or not to stop supply of the operation clock to the transmission unit based on the network information; a setting unit that sets, in a case where it is determined by the determination unit to stop supply of the operation clock to the transmission unit, a first period in which supply of the operation clock is stopped based on the network information; and a clock control unit that stops supply of the operation clock to the transmission unit in the first period and, after the first period has expired, resumes supply of the operation clock to the transmission unit, wherein the determination unit calculates, based on the network information, a number of first cycles from when the arithmetic unit processes one input data until when the arithmetic unit requests for the next input data and, in a case where the number of first cycles is larger than a number of second cycles that is a sum of the number of cycles for stop of clock supply to the transmission unit by the clock control unit and the number of cycles for resumption of data transmission by resumption of clock supply, determines to stop supply of the operation clock to the transmission unit.
2. The information processing apparatus according to claim 1, wherein the setting unit sets the first period after transition is made to a state in which transmission of the input data to the arithmetic unit is not used or expected.
3. The information processing apparatus according to claim 1, wherein the setting unit determines as the

first period the number of cycles obtained by subtracting the number of second cycles from the number of first cycles.
4. An information processing apparatus comprising: one or more memories storing instructions; and one or more processors executing the instructions to cause the information processing apparatus to function as: an arithmetic unit that performs arithmetic operation processing using a hierarchical network; a storage unit that stores input data inputted to the arithmetic unit and output data outputted from the arithmetic unit; a transmission unit that transmits to the arithmetic unit the input data stored in the storage unit; a reception unit that receives and stores in the storage unit the output data from the arithmetic unit; and a control unit that controls, in a case where it is not possible to transfer the output data, supply of an operation clock to the reception unit based on network information that indicates a structure of the hierarchical network, wherein the hierarchical network is a CNN (Convolutional Neural Network), wherein the arithmetic operation processing includes convolution operation processing on the input data, the network information includes at least one of a size of a convolution kernel that is used in the convolution operation processing on the input data and the number of feature planes, wherein the control unit includes: a determination unit that determines, in a case where it is not possible to transfer the input data, whether or not to stop supply of the operation clock to the reception unit based on the network information; a setting unit that sets, in a case where it is determined by the determination unit to stop supply of the operation clock to the transmission unit, a second period in which supply of the operation clock is stopped based on the network information; and a clock control unit that stops supply of the operation clock to the reception unit in the second period and, after the second period has expired, resume supply of the operation clock to the reception unit, wherein the determination unit calculates, based on the network information, a number of third cycles from when the arithmetic unit outputs one output data until when the arithmetic unit outputs the next output data and, in a case where the number of third cycles is larger than a number of fourth cycles that is a sum of the number of cycles for stop of clock supply to the reception unit by the clock control unit and the number of cycles for resumption of data reception by resumption of clock supply, determines to stop supply of the operation clock to the reception unit.
5. The information processing apparatus according to claim 4, wherein the setting unit sets the second period after transition is made to a state in which reception of output data from the arithmetic unit is not used or expected.
6. The information processing apparatus according to claim 4, wherein the setting unit determines as the second period the number of cycles obtained by subtracting the number of fourth cycles from the number of third cycles.
7. A method of controlling an information processing apparatus comprising an arithmetic unit, a storage unit, a transmission unit, and a reception unit, the method comprising: transmitting input data stored in the storage unit to the arithmetic unit by the transmission unit; performing arithmetic operation processing on the transmitted input data using a hierarchical network by the arithmetic unit; and receiving the output data from the arithmetic unit and storing the received data in the storage unit by the reception unit, and the method further comprising: determining whether it is possible to transmit the input data from the storage unit to the arithmetic unit; and in a case where it determined that it is not possible to transmit the input data, a control unit for controlling supply of an operation clock to the transmission unit based on network information that indicates a structure of the hierarchical network, wherein the hierarchical network is a CNN (Convolutional Neural Network), wherein the arithmetic operation processing includes convolution operation processing on the input data, the network information includes at least one of a size of a convolution kernel that is used in the convolution operation processing on the input data and the number of feature planes, wherein the control unit includes: a determination unit that determines, in a case where it is not possible to transfer the input data, whether or not to stop supply of the operation clock to the reception unit based on the network information; a setting unit that sets, in a case where it is determined by the determination unit to stop supply of the operation clock to the transmission unit, a second period in which supply of the operation clock is stopped based on the network information; and a clock control unit that stops supply of the operation clock to the reception unit in the second period and, after the second period has expired, resume supply of the operation clock to the reception unit, wherein the determination unit calculates, based on the network information, a number of third cycles from when the arithmetic unit outputs one output data until when the arithmetic unit outputs the next output data and, in a case where the number of third cycles is larger than a number of fourth cycles that is a sum of the number of cycles for

stop of clock supply to the reception unit by the clock control unit and the number of cycles for resumption of data reception by resumption of clock supply, determines to stop supply of the operation clock to the reception unit.

8. A method of controlling an information processing apparatus comprising an arithmetic unit, a storage unit, a transmission unit, and a reception unit, the method comprising: transmitting input data stored in the storage unit to the arithmetic unit by the transmission unit; performing arithmetic operation processing on the transmitted input data using a hierarchical network by the arithmetic unit; and receiving the output data from the arithmetic unit and storing the received data in the storage unit by the reception unit, and the method further comprising: determining whether it is possible to receive the output data from the arithmetic unit to the storage unit; and in a case where it determined that it is not possible to transmit the input data, a control unit for controlling supply of an operation clock to the transmission unit based on network information that indicates a structure of the hierarchical network, wherein the hierarchical network is a CNN (Convolutional Neural Network), wherein the arithmetic operation processing includes convolution operation processing on the input data, the network information includes at least one of a size of a convolution kernel that is used in the convolution operation processing on the input data and the number of feature planes, wherein the control unit includes: a determination unit that determines, in a case where it is not possible to transfer the input data, whether or not to stop supply of the operation clock to the reception unit based on the network information; a setting unit that sets, in a case where it is determined by the determination unit to stop supply of the operation clock to the transmission unit, a second period in which supply of the operation clock is stopped based on the network information; and a clock control unit that stops supply of the operation clock to the reception unit in the second period and, after the second period has expired, resume supply of the operation clock to the reception unit, wherein the determination unit calculates, based on the network information, a number of third cycles from when the arithmetic unit outputs one output data until when the arithmetic unit outputs the next output data and, in a case where the number of third cycles is larger than a number of fourth cycles that is a sum of the number of cycles for stop of clock supply to the reception unit by the clock control unit and the number of cycles for resumption of data reception by resumption of clock supply, determines to stop supply of the operation clock to the reception unit.